

Swaption model		1	into	3		
Time	Swap rate	zc	Period	DF(0, t)		
1	5.000%	5.000%	0x1		d_1	-0.4496
2	5.100%	5.103%	1x2	0.9053	d_2	-0.6096
3	5.200%	5.207%	2x3	0.8587		
4	5.300%	5.314%	3x4	0.8129	$FN(d_1) - XN(d_2)$	0.001702
5	5.400%	5.423%	4x5		Payer swaption (in bps)	1,810,044 43.9
F	5.411%		$A =$	2.5770		
σ	16.0%					
T	1		Notional	412,571,222		
X	5.889%					

Swaption model		1	into	1		
Time	Swap rate	zc	Period	DF(0, t)		
1	5.000%	5.000%	0x1		d_1	-0.4264
2	5.100%	5.103%	1x2	0.9053	d_2	-0.6064
3	5.200%	5.207%	2x3			
4	5.300%	5.314%	3x4		$FN(d_1) - XN(d_2)$	0.001888
5	5.400%	5.423%	4x5		Payer swaption (in bps)	1,996,608 17.1
F	5.205%		$A =$	0.9053		
σ	18.0%					
T	1		Notional	1,167,982,789		
X	5.712%					

Figure S.26: Mark-to-Market of Bearish Steepener After One Year

b) As shown in Figure S.26, we have hit our stop-loss. The value of the 1 into 3 payer swaption that we are long is \$1,810,044, and the value of the 1 into 1 payer swaption that we are short is \$1,996,608 for a mark-to-market loss of \$186,564.

c) Although the differential in current forward rates is advantageous compared to the strikes, this effect has been dominated by the adverse move in vols.

Chapter 9

1. a) First, consider the case in which the 4-year swap rate, one year forward and 1-year LIBOR, one year forward are negatively correlated. Then as 1-year LIBOR, one year forward goes down (in which case the chances of the swap knocking out increase), the 4-year swap rate, one year forward will tend to go up. As the 4-year swap rate, one year forward goes up, the value of the residual 4-year swap in a year decreases to the fixed rate receiver. Thus, the swap will tend to knock out in situations in which the remaining 4-year swap is decreasing in value to the client.

Now consider the case in which the 4-year swap rate, one year forward and 1-year LIBOR, one year forward are positively correlated. Then as 1-year LIBOR, one year forward goes down (in which case the chances of the swap knocking out increase), the 4-year swap rate, one year forward will tend to go down as well. But as the 4-year swap rate, one year forward goes down, the value of the residual 4-year swap in a year increases to the fixed rate receiver. Thus, the swap will tend to knock out in situations in which the remaining 4-year swap is increasing in value to the client.

In summary, when the correlation is negative, the knockout feature will tend to knock out the swap in situations in which it may be advantageous from the point of view of the fixed rate receiver. But when the correlation is positive, the knockout feature will tend to knock out the swap in situations in which it is disadvantageous from the point of view of the fixed rate receiver. Thus, everything else the same, positive correlation hurts the fixed rate receiver and he is compensated for this by receiving a higher fixed coupon.

In the USD swaps market, it is often empirically observed that short-term LIBOR forwards are positively correlated with longer-term forward swap rates. Given this fact, knockout swaps such as the one that we have considered here are typically priced assuming positive correlation between the forward LIBOR rate that determines whether or not the swap knocks out

and the residual forward swap rate. Making the assumption of positive correlation serves to increase the fixed rate that can be paid in the trade.

b) Yes, this is possible. The 4-year swap rate, one year forward shares, in its construction, a LIBOR forward in common with 1-year LIBOR, one year forward. This fact in isolation points to positive correlation between the two rates. However, the 4-year swap rate, one year forward is constructed with an additional three forward LIBOR rates. It is possible that these three other forward rates are negatively correlated with the front forward in such a way that the 1x5 and 1x2 rates are uncorrelated.

In fact, we saw such a thing (in which near term LIBOR forwards were negatively correlated with longer term LIBOR forwards) during certain periods of the financial crisis. There were times during the financial crisis when we saw long-term rates (for example, the 10-year swap rate) decrease significantly. At the same time, we saw short-term LIBOR rates rise as financial institutions were less willing to lend to one another. This was a time during which short-term LIBOR forwards were negatively correlated with long-term LIBOR forwards.

Appendix A

1. a) The value of the European call is 12.5, as shown in Figure S.27. In this figure, h and B represent the amount of shares and borrowing, respectively, in the replicating portfolio. The quantities c_{1u} and c_{1d} represent the value of the call at time $t = 1$ in the upper node and lower node, respectively, and c is the value of the call today.

b) The value of the European put is 3, as shown in Figure S.28. This figure also shows the appropriate replicating portfolio.

c) From put-call parity, $p = c - S + PV(X)$. Thus, the put should equal $12.5 - 32 + 40/(1.33)^2 = 3$, which it does.

d) The value of the American call is the same as the value of the European call, since there is no value to early exercise.

e) If at time $t = 1$ we reach the lower node, the unexercised put is worth 14, although immediate exercise provides a payoff of 24. Thus, early exercise is optimal at $t = 1$ at the lower node (but not at the upper node). The value of the American put at time $t = 0$ if it is not exercised is therefore

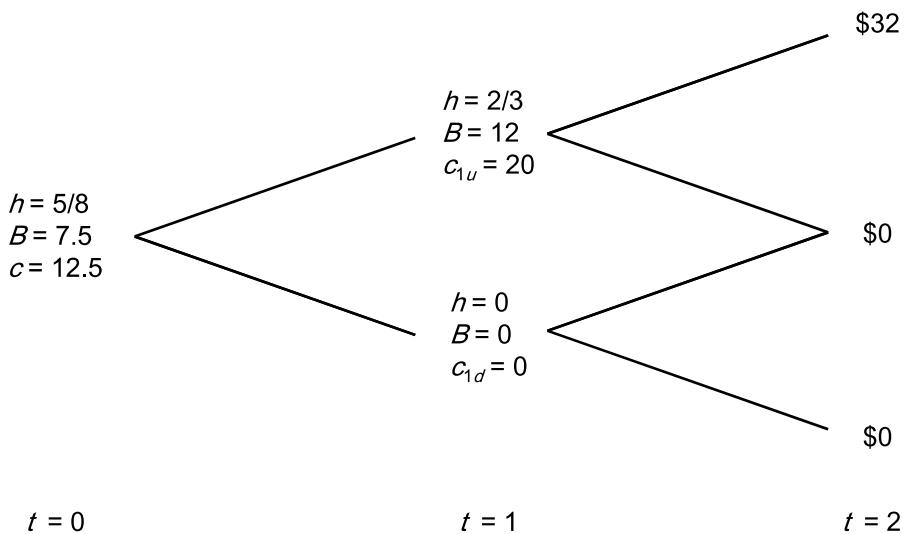


Figure S.27: Solving for the Value of the Call

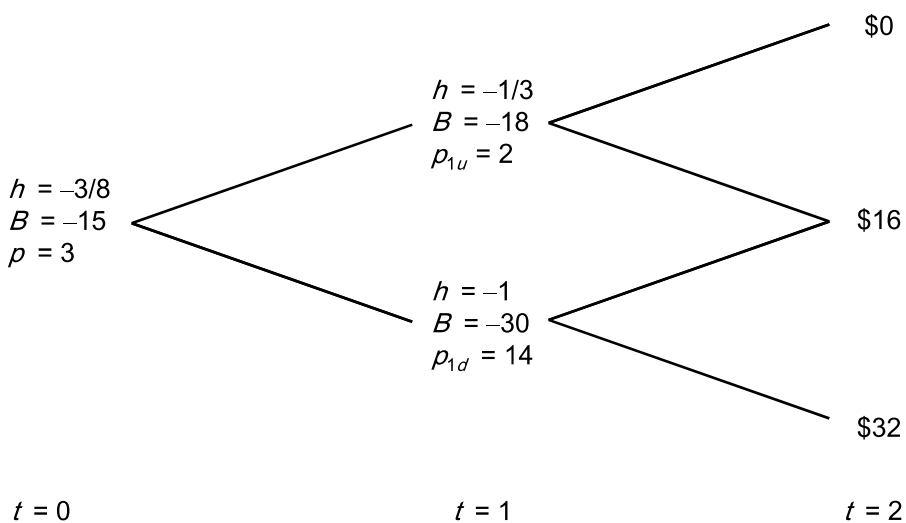
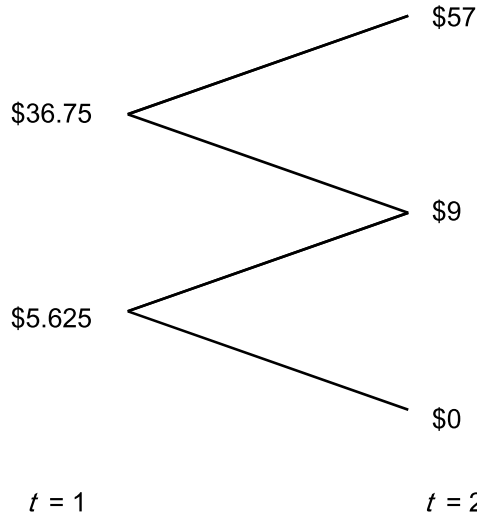


Figure S.28: Solving for the Value of the Put

Figure S.29: Value of the Underlying Call Option at Time $t = 1$

equal to $(3/4) \times [(5/6) \times 2 + (1/6) \times 24] = 4.25$. However, if at time $t = 0$, the put is exercised immediately, the payoff is 8, which exceeds the value if unexercised. The value of the American put is therefore 8.

2. At time $t = 1$, the holder of the compound option will choose to exercise and deliver the call option if and only if the value of the call option is less than \$20. Figure S.29 shows the value of the call option at each possible node at time $t = 1$. Thus, the holder will only exercise if we reach the bottom node at time $t = 1$. In this case, the value exercising is the difference between the \$20 received by exercising and the \$5.625 value of the delivered call, or \$14.375. Thus, the value at time $t = 0$ of the compound option is $(3/4) \times [0 \times (5/6) + 14.375 \times (1/6)] = 1.797$.

4. Referring to Figure S.30, we find that $u = 1.2$, $d = 0.2$, and since $r = 0.08$, we therefore have $p_n = 0.88$ from Equation A.16. The value of the derivative contract is therefore $(1/1.08) \times (0.88 \times 20 + 0.12 \times 0) = 16.30$. Therefore, choose b).

7. The arbitrage strategy and cash flows are shown in Table S.2

To avoid arbitrage, we need $2c_2 - c_1 - c_3 \leq 0$, which yields the desired result.

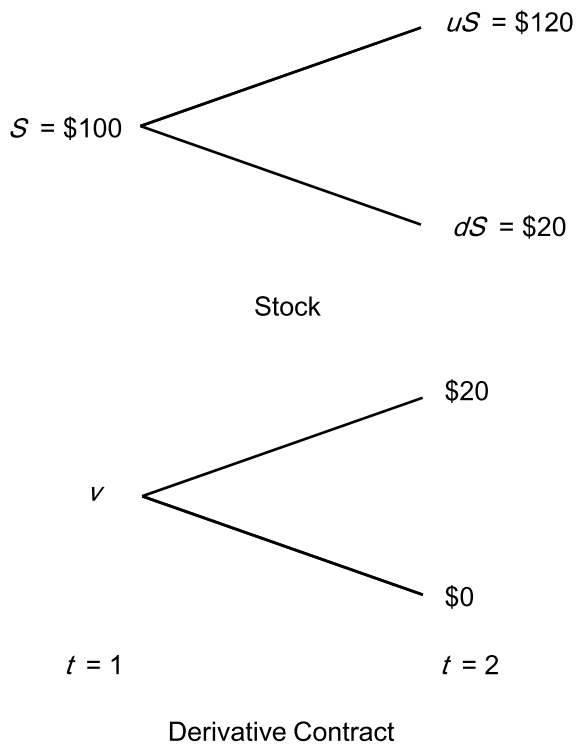


Figure S.30: The Derivative Contract

Action Today	Cash Flow Today	Cash Flows at Expiry			
		$S_T < X_1$	$X_1 \leq S_T < X_2$	$X_2 \leq S_T \leq X_3$	$X_3 < S_T$
Buy c_1	$-c_1$	0	$S_T - X_1$	$S_T - X_1$	$S_T - X_1$
Buy c_3	$-c_3$	0	0	0	$S_T - X_3$
Short $2c_2$	$+2c_2$	0	0	$-2(S_T - X_2)$	$-2(S_T - X_2)$
Net	$+2c_2 - c_1 - c_3$	0	$S_T - X_1 \geq 0$	$2X_2 - S_T - X_1$ $= X_3 - S_T \geq 0$	0

Table S.2: Arbitrage Table

10. Buy short-dated calls and sell long-dated calls, both near-the-money.

11. No. Suppose both stocks are lognormally distributed. Suppose both options have six months to expiry, and the risk-free rate is 10%. If one stock is at \$10 and its vol is 20%, then the call on this stock is worth \$0.83, according to Black-Scholes. If the second stock is at \$100 and its vol is 10%, then the call on this stock is worth \$5.85, according to Black-Scholes.

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